Software Requirements Specification (SRS)

GeoPulse : Indoor Outdoor Positioning System

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1 Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for an indoor positioning and navigation system designed specifically for visually impaired users [cite: 375-376]. This system utilizes a simulated beacon network, smartphone IMU sensors, and quantum-optimized routing to provide highly accurate, real-time spatial audio guidance[cite: 377].

1.2 Scope

The system will simulate an indoor beacon network using a cluster of stationary smartphones broadcasting BLE signals [cite: 378-379]. A client application will ingest these signals alongside IMU data to pinpoint the user's location via sensor fusion[cite: 380]. The system includes an SOS fall detection mechanism and integrates with a cloud-based Quantum Processing Unit (QPU)[cite: 381].

2 Overall Description

2.1 User Characteristics

The primary users are visually impaired individuals requiring autonomous indoor navigation [cite: 383-384]. The interface must be entirely non-visual, relying on multi-lingual voice commands and 3D spatial audio feedback[cite: 385].

2.2 Operating Environment

- **Client App:** Android/iOS mobile devices with active Bluetooth receivers, compass, and accelerometer sensors[cite: 387].
- **Backend Server:** Python 3.10 environment utilizing Flask[cite: 388].
- **External APIs:** IBM Qiskit API for quantum route optimization[cite: 389].

3 System Architecture & Models

This section outlines the logical and data flow structures of the application [cite: 391-392].

3.1 System Architecture

The system utilizes a "Dumb Client / Smart Backend" architecture to minimize latency on the mobile device while performing heavy sensor fusion and API routing on the server [cite: 394-395].

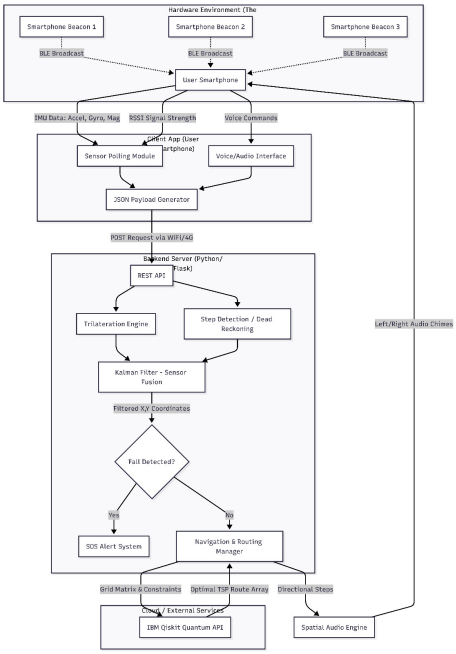

Figure 1: High-Level System Architecture

3.2 Use Case Model

The Use Case model details user interactions and system boundaries [cite: 397-398].

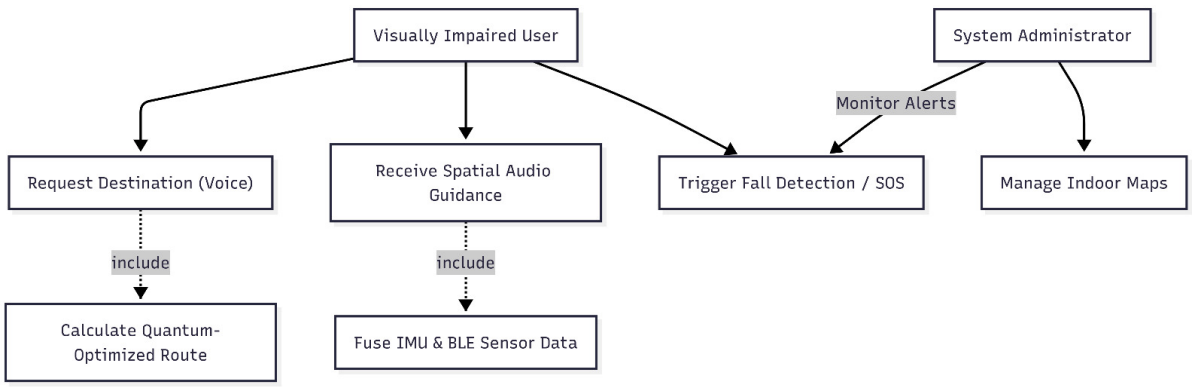

Figure 2: User Interactions and System Boundaries

3.3 Data Flow Diagram (DFD)

The DFD illustrates how raw sensor data is ingested, processed via sensor fusion, optimized using quantum algorithms, and converted into audio instructions [cite: 411-412].

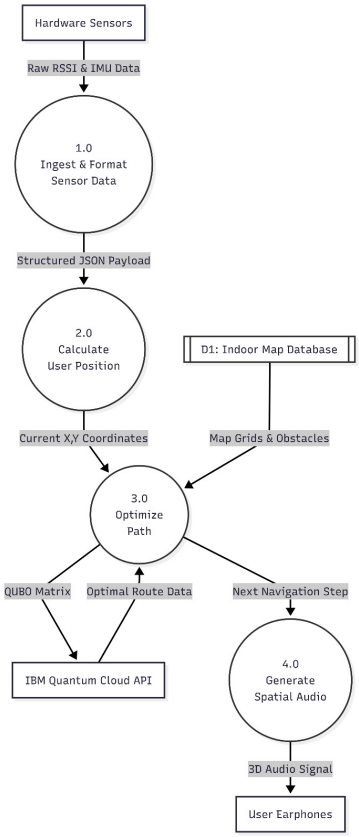

Figure 3: Level 1 Data Flow: Sensor Ingestion to Audio Output

3.4 Activity Diagram

The Activity diagram outlines the continuous real-time navigation and decision-making loop handled by the system brain [cite: 426-427].

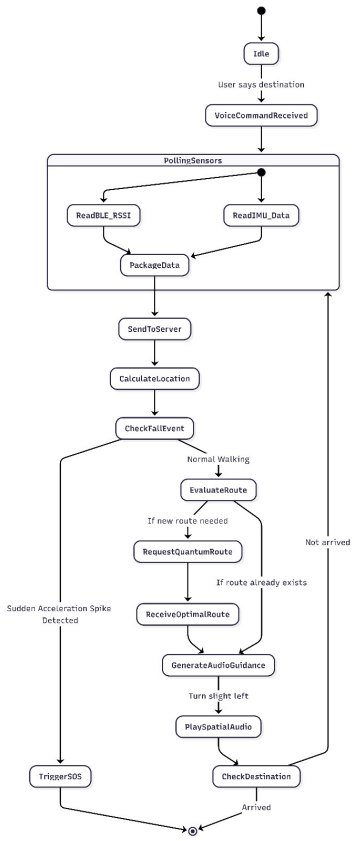

Figure 4: Real-Time Navigation and Decision Loop

4 System Features

4.1 BLE Trilateration Engine

The system must parse incoming RSSI signals from at least three known beacon nodes to calculate a rough geometric coordinate mapping [cite: 429-430].

4.2 IMU Sensor Fusion (Pedestrian Dead Reckoning)

The backend must utilize an algorithm (e.g., Kalman Filter) to fuse bouncy BLE data with step-detection data from the accelerometer for smooth location vectors [cite: 431-432].

4.3 Quantum Route Optimization

The system will map the floor plan into a QUBO matrix and query the IBM Qiskit cloud to return mathematically shortest, safest paths [cite: 433-434].

4.4 Fall Detection & SOS Trigger

The client application must continuously monitor the accelerometer for sudden G-force spikes indicative of a fall and trigger an automated SOS alert [cite: 435-437].

5 Non-Functional Requirements

- **Latency:** End-to-end data processing must occur in under 500ms to ensure user safety [cite: 457-458].
- **Availability:** If a single node fails, the system must degrade gracefully using pure IMU Dead Reckoning[cite: 459].

6 References

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